

Empiricist

Smoke cards · GPU honesty · directional success only

Label recoveries by disturbing the expert, not by deploying the learner.

49/79/72

paper HSR

0-8 GB

Hopper smoke

Humanoid=N

200-iter defer

HSR=N

real robot

directional $\neq$ 79%

success bar

Hypotheses (falsifiable)

Pass = directional confirmation + α/Σ logs — not paper parity

H-dart>bc

DART($\alpha \approx 1-3$) > BC on Hopper/Walker return (matched demos/arch)

H-opt>iso

Optimized Σ > isotropic $\Sigma=I$ at same intended noise scale

H-alpha-sweet

α has interior optimum; too large hurts collect-return / final policy

H-parity-dagger

DART return $\approx$ DAgger; DART $\downarrow$ wall $\cdot$ $\uparrow$ supervisor collect-return

H-shift

DART shrinks rollout–demo surrogate gap faster than BC (Fig.4)

H-recover

Under mid-traj kicks, DART recovers $\geq$ BC

Minimal matrix (course smoke)

BC | isotropic | DART_α × Hopper × K=2

DEFAULT MATRIX

{method: BC | isotropic | DART_α ∈ {1,3} | (opt) DAgger_β=0.5} × {env: Hopper} × {N_demos: 10-20} × {K: 2}

+ Tr(Σ) ∈ {0.005, 0.5, 5} one-factor card · log α, tr(Σ), supervisor return@collect, wall, train/rollout loss gap

P0a α card

Σ then α ∈ {1,3,6} × T tr(Σ)
mirror HSR without HSR

P0b type card

isotropic Σ=I vs DART Σ
same budget

P0c Tr card

Wishart/scaled random Σ
{0.005,0.5,5} vs DART

GPU / RAM / TIME honesty (DELIVERABLE)

Paper ran on 2017 MacBook CPU — do not over-provision

Workload	VRAM	Wall	Smoke
Hopper BC/DART ≤ 20 demos K=2	0-8 GB	30-90 min	Y
α sweep $\times 4$	0-8 GB	2-6 h	Y
Dagger vs DART lite	0-8 GB	2-8 h	Y
Recovery-kick eval	0-4 GB	minutes	Y
Walker / Half-Cheetah scale	0-8 GB	4-16 h	Partial
Offline noise $\rightarrow$ DP/ACT FT	12-24 GB	8-24 h	Partial
Humanoid-200 paper protocol	CPU-days+	$\gg 1$ day	N
Real HSR $\times 4$ humans	—	robot+IRB	N

P0 cards — α / $\text{Tr}(\Sigma)$ / isotropic

Ablation Bot #1-#3 proxies

α magnitude

Sweep $\alpha \in \{1, 3, 6\}$
Expect inverted-U
(HSR moral: $3 > 6$)
Metric: return +
supervisor@collect
Smoke: Y · 1-6 GPU-h

opt vs isotropic

DART $\Sigma + \alpha$ vs $\Sigma = I$
Expect iso weak/unsafe
Metric: return, fall rate
Smoke: Y · 1-4 GPU-h

$\text{Tr}(\Sigma)$ Wishart

$\text{Tr} \in \{0.005, 0.5, 5\}$
 $\text{tr} \approx 0.5$ can match DART
wrong mag hurts
Smoke: Y · 2-6 GPU-h

Success bar = directional NOT paper parity

Do not claim Humanoid 3x or HSR 79% from Hopper smoke

DART reduces shift vs BC

↑ return and/or ↓ rollout—demo L2 on Hopper

NOT: Exact Fig.2 curves

DART $\approx$ DAgger

Returns within noise on ONE loco env

NOT: Humanoid parity

Cheaper / safer collect

↓ wall or ↑ supervisor@collect vs DAgger

NOT: Exact 3x / -5% / -80%

Isotropic insufficient

DART > isotropic at matched scale

NOT: Paper “unsafe” qualitative

α matters

Interior α beats extremes

NOT: HSR 79%/72%

Recovery demos help

Higher recovery SR under kicks

NOT: Real clutter recoveries

Motion / covariate-shift deep dive (required)

What to measure when claiming H-shift / H-recover

Compounding over T

Log dual-dist surrogate loss (supervisor vs robot) vs iteration — Fig.4 moral.

Recovery boundary

Script mid-traj kicks; measure restabilization SR / return-after-kick.

Action $\neq$ state noise

Factorial: control-stream noise vs obs dropout vs domain rand — separate axes.

Schedule

Log α , $\text{tr}(\Sigma)$, β every round; wrong scale falsifies H-opt and H-alpha.

Vs DAgger locus

Same shift objective; log who acted + collect reward — safety metric first-class.

ACT fine-teleop claim test

Chunking + noise — test, don't assume

ACT says: DART-style noise yields corrective behavior but can break fine teleop → prefer chunking.

Empiricist test: compare chunking-only vs noise-only vs chunking+moderate- α on a precision insert proxy.

Noise-only

Expect: recovery $\uparrow$ on loco
Expect: precision SR $\downarrow$
if α ungated

Chunking-only

Expect: horizon $\uparrow$
May still lack recovery
labels off-manifold

Chunk + gated α

Hypothesis: best of both
if noise task-gated
(loco $\uparrow$ / contact $\downarrow$)

Paper targets (not smoke requirements)

Fig.2 / Fig.3 / App Fig.4–5 — directional morals only

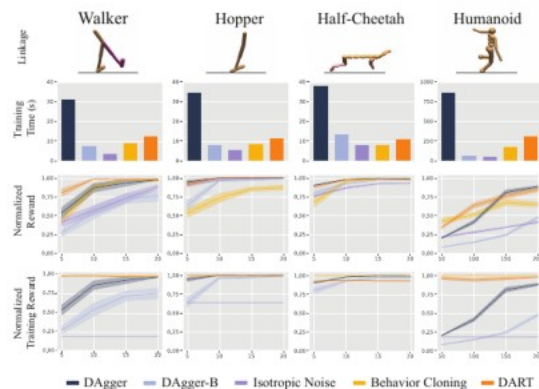


Figure 2: Top: The four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. Top Middle: The time, in seconds, to achieve the performance level reported below. DART achieves similar performance to DAgger in all 4 domains, but requires significantly less computation because it doesn't require retraining the current robot policy after each demonstration. DAgger-B reduces the computation required by less frequently training the robot, but suffers significantly in performance in domains like Humanoid. Bottom Middle: The learning curve with respect to reward obtained during training with each algorithm plotted across the number of demonstrations. Bottom: The reward obtained during collection of demonstrations for learning. DART receives near the supervisor's reward at all iterations whereas DAgger can be substantially worse in the beginning.

policies in Tensorflow. At each iteration we collected one demonstration. In order to make the task challenging for Imitation Learning, we used the same technique as in [6], which is to sub-sample 50 state and control pairs from the demonstrated trajectories, making the learners receive less data at each iteration.

We compare the following five techniques: Behavior Cloning, where data is collected from the supervisor's distribution without noise injected; DAgger, which is an on-policy method that stochastically samples from the robot's distribution and then fully retrains the robot's policy after every demonstration is collected; DAgger-B, which is DAgger but the retraining is done intermittently to reduce computation time; Isotropic-Gaussian which injects non-optimized isotropic noise; and DART. See supplement material for how the hyper-parameters for each method are set.

We used the following MuJoCo environments: Walker, Hopper, Humanoid, and Half-Cheetah. The size of the state and control space for each task is $\{|X| = 17, |U| = 6\}$, $\{|X| = 11, |U| = 3\}$, $\{|X| = 376, |U| = 17\}$, $\{|X| = 117, |U| = 6\}$, respectively. All experiments were run on a MacBook Pro with an 2.5 GHz Intel Core i7 CPU. We measured the cumulative reward of each learned policy by rolling it for 500 timesteps, the total computation time for each method and the cumulative reward obtained during learning.

Fig. 2 shows the results. In all domains, DART achieves parity with DAgger, whereas Behavior Cloning and DAgger-B are below this performance level in Walker and Humanoid. For Humanoid, DART is 3x faster in computation time and during training only decreases the supervisor's cumulative reward by 5%, whereas DAgger executes policies that have over 80% less cumulative reward than the supervisor. The reason for this is that DAgger requires constantly updating the current robot policy and forces the robot into sub-optimal states. While, one way to reduce this computation is to decrease



Figure 3: Left: Experimental setup for the grasping in clutter task. A Toyota HSR robot uses a head-mounted RGBD camera and its arm to push obstacle objects out of the way to reach the goal object, a mustard bottle. The robot's policy for pushing objects away uses a CNN trained on images taken from the robot's Primesense camera, an example image from the robot's viewpoint is shown in the orange box. Right: The Success Rate for Behavior Cloning, DART($\alpha = 3$) and DART($\alpha = 6$). DART($\alpha = 3$) achieves the largest success rate.

the number of times the policy is updated. DAgger-B illustrates that this can significantly deteriorate performance. Lastly, naively applying isotropic noise does not perform well, and leads to unsafe policies during execution, which suggests the need for optimizing the level of noise.

5.2 Robotic Grasping in Clutter

We evaluate DART with human supervisors in a grasping in clutter task on a Toyota HSR robot. We consider a task inspired by the Amazon Picking Challenge [2]. The goal of the task is for the robot to retrieve a goal object from a cupboard. The task is challenging because the goal object is occluded by obstacle objects and the robot must reason about how to clear a path based on observations taken from an eye-in-hand perspective. The objects are 6 common household food items, which consist of boxes and bottles with varying textures and mass distributions. The target object is fixed to always be a mustard bottle. The robot, task and image viewpoint are shown in Fig. 3. See supplement material for additional information on the task.

We use 4 supervisors who have robotics experience but not specifically in the field of Imitation Learning and compare Behavior Cloning and DART. When performing the study, we first collect $N = 10$ demonstrations with Behavior Cloning (i.e. no noise) and then in a counter-balanced order collect $N = 30$ more demonstrations with each technique. Our experiment was within-subject, have every supervisor perform all three methods.

We only updated the noise parameter after the first 10 demonstrations (i.e. $K = 2$). The final robot policy was then trained on the total of 40 demonstrations. We consider two different choices of α : $\alpha = 3\text{Tr}(\Sigma_1)$ and $\alpha = 3\text{Tr}(\Sigma_1)$. These choices correspond to the intuition that with small datasets in high dimensional image space the robot will have significant shift from the current loss on the supervisor's distribution. Through the rest of the paper, we will write our choices as $\alpha = 3$ and $\alpha = 6$ for brevity in notation.

During policy evaluation, we measured success as 1) the robot is able to identify the goal object and 2) there being a clear path between its gripper and the object. Once these conditions are identified, as detailed in the supplement material, an open-loop motion plan is generated to execute a grasp around the target object.

In Fig. 3, we report the average performance of the three techniques across DART 20 times on different sampled initial states. DART with $\alpha = 3$ performs the best out of the three techniques, with a 79% success rate over traditional Behavior Cloning's 49% success rate. Interestingly, DART with $\alpha = 6$ performs better than Behavior Cloning, but only has a 72% success rate. This may suggest this level of noise was potentially too high for the human supervisor.

6 Conclusion

This paper considers injecting noise into the supervisor's demonstrations to minimize covariate shift. Our algorithm, DART, provides the robot with corrective examples at the boundary of the supervisor's policy. By being concentrated around the supervisor's policy, it collects demonstrations without visiting highly sub-optimal states and is easier for human supervisors tele-operating. We demonstrate it achieves parity with on-policy methods in simulated MuJoCo domains and is significantly better than traditional Behavior Cloning on a real robot. For code and more information see <http://berkeleyautomation.github.io/DART/>

App Fig.4-5 — shift gap + $\text{Tr}(\Sigma)$

Magnitude dominates if scale wrong

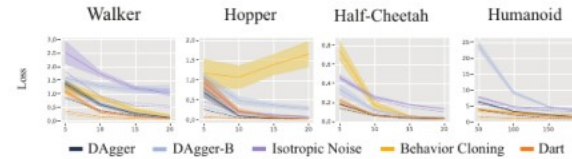


Figure 4: The covariate shift on the four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. The dashed lines correspond to the surrogate losses on the supervisor distributions, $E_{p(\xi) \sim p_{\text{sup}}} J(\theta^H, \theta^* | \xi)$, and the solid lines correspond to the loss on the robot distributions $E_{p(\xi) \sim p_{\text{rob}}} J(\theta^H, \theta^* | \xi)$. DART is able to reduce the covariate shift in all four domains, which is illustrated by the dash line converging to the solid line. Additionally, DAgger is also able to reduce the shift. However, Behavior Cloning is less effective at doing so, which is illustrated by a slower convergence rate.

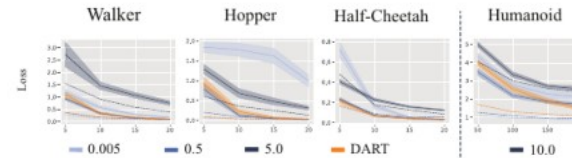


Figure 5: The losses of learners trained from supervisors with randomly sampled covariance matrices are compared against the losses of DART. Again, the dashed lines represent the loss on the supervisor distributions while the solid lines represent the loss on the robot distributions. When the simulated error is well-chosen, such as $\text{Tr}(\Sigma) = 0.5$ in these experiments, the performance matches that of DART and the covariate shift is reduced. However higher or lower levels of the noise can cause drastically higher losses.

open-loop motion plan is generated to execute a grasp around the target object. In order for the robot to identify the target object, we use the Faster R-CNN trained on the PASCAL VOC 2012 dataset, which has a bottle object category [12]. Once the mustard bottle has been identified and the a path is cleared a human supervisor tells the robot to execute the open-loop motion plan towards a fix position that the goal object is at. The human supervisor is used to ensure that the success criteria is not prone to error in the system, but the supervisor is blinded to which policy is being evaluated to prevent biases. We evaluate the policy 20 times on different sampled initial states. A video of the learned policy and the robot setup can be found here: <https://youtu.be/LfMD6911esg>.

Knobs to pin on every smoke card

From App §4.3 + DELIVERABLE

Dagger β

0.5 (paper best)

MuJoCo α

$T \text{tr}(\Sigma_k)$ — no extra prior

HSR α

$\{3,6\} \times T \text{tr}(\Sigma_1)$ — 3 wins

Isotropic

$\Sigma=I$ always — expect fail

Σ source

Held-out prior-iter demos

Subsample

50 (s,a)/traj poverty trick

K

HSR K=2; MuJoCo few updates

Loss

L2 controls; theory $l \in [0,1]$

Defer / N — do not launch from quiet box

Honesty beats cargo-cult Humanoid-200

N

Full Humanoid-200 + paper TRPO supervisor retrain

N

Real Toyota HSR clutter × 4 human supervisors

N

Physical DAgger on human task (paper never ran)

Partial

Walker/HalfCheetah scale-up after Hopper pass

Partial

Offline DART-lite action-noise on DP/ACT demos

Y first

Hopper P0: BC vs iso vs DART(α) + Tr sweep

Tattoo card

HSR 49% / 79% / 72% · +62% rel · Humanoid 3× wall-clock · collect −5% (DART) vs −80% (DAgger) · α inverted-U

49%

BC

79%

$\alpha=3$

72%

$\alpha=6$

+62%

rel

3×

wall

−5% vs −80%

collect

0-8 GB

Hopper smoke Y

HSR/Hum = N

defer

Close — Empiricist

P0 first · directional only · $\log \alpha/\Sigma$

1. Run Hopper matrix: BC | isotropic | DART_ $\alpha \in \{1,3\}$ | K=2 before anything else.

2. Pass = directional H-dart>bc or H-opt>iso or H-parity compute/safety + logs.

3. Never claim Humanoid 3x or HSR 79% from smoke.

4. Test ACT claim: chunking $\pm$ gated noise on a precision proxy.